

Chapter 4: Derivability

① Definition:

Let I be an open interval of $\mathbb{R}$ and:

$f: I \longrightarrow R$ be a function

We say that f is differentiable at x_0 when there exist a finite number at x_0 such that.

$$\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$$

Note A: If f is diff at every point $x \in I$ then f is diff at I and the function

is the derivative function of f .

Notes

f : differentiable $\xrightarrow{\checkmark}$ Continuous
 $\xleftarrow{\times}$

↓
example: the function $f(x) = |x|$

continuous over $\mathbb{R}$ $\swarrow \searrow$ not diff over $\mathbb{R}$.

② proposition (chain Rule):

$$g: I \longrightarrow \mathcal{I} \text{ (diff at } u_0)$$
$$f: \mathcal{I} \longrightarrow \mathbb{R} \text{ (diff at } g(x_0))$$

↳ $f \circ g$ is diff at no and:

$$(f \circ g)'(x_0) = f'(g(x_0)) \times g'(x_0)$$

③ Corollary:

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① $f: I \rightarrow J$ (bijective and diff function)

$$f^{-1}: J \rightarrow \mathbb{R} \quad (f^{-1} \text{ is diff}^p \text{ function})$$

② $f'(x) \neq 0$, $\forall x \in I$ and $\forall x \in J$ we have:

$$[f^{-1}(x)]' = \frac{1}{f'(f^{-1}(x))}$$

④ Rolle Theorem :

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Let $f: [a, b] \rightarrow \mathbb{R}$ s.t.

- ① f is continuous $[a, b]$
- ② f is diff at $[a, b]$
- ③ $f(a) = f(b)$

Then, $\exists c \in]a; b[\text{ s.t. : } f'(c) = 0$
 $= c$ here called: "the critical point"

↳ How to find c?

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we solve the equation $f'(c) = 0$

⑤ Mean value Theorem:

Let $f: [a; b] \rightarrow \mathbb{R} \leq \mathbb{T}$ $\left\{ \begin{array}{l} \textcircled{1} f \text{ is Cont over } [a; b] \\ \textcircled{2} f \text{ is diff over } [a; b] \end{array} \right.$

Then:

$$f(b) - f(a) = f'(c)(b-a)$$

↳ How to use TIVT?

when the exo asks to prove an equation
(check the course examples)

Corollary: (Inequality of mean value Thm)

Let $f: I \rightarrow \mathbb{R}$ be a continuous diff funct. If:

$$\exists \pi > 0 \text{ s.t. } \forall x \in I: |f'(x)| \leq \pi$$

Then: $|f(x) - f(y)| \leq \pi |x - y|$
 $\forall x, y \in I$

Example: prove that: $\forall x, y \in \mathbb{R}$

$$|\sin x - \sin y| \leq |x - y|$$

Let $f(x) = \sin x$.

$x \rightarrow \sin x$ is diff function over $\mathbb{R}$.

Then $x \rightarrow \sin x$ is cont function over $\mathbb{R}$

$$f'(x) = \cos x$$

$$|f'(x)| = |\cos x| \leq 1$$

$$\exists \pi = 1 > 0 \text{ s.t. } |\cos x| \leq 1$$

Then, by mean value theorem, $\forall x, y \in \mathbb{R}$

$$|f(x) - f(y)| \leq \pi |x - y|$$

$$|\sin x - \sin y| \leq 1 |x - y|$$

Hence: $|\sin x - \sin y| \leq |x - y|$

① Hospital Rule:

Let $f, g: I \rightarrow \mathbb{R}$ be two diff funct

and let $x_0 \in I$

we assume that $\left\{ \begin{array}{l} \textcircled{1} f'(x_0) = g'(x_0) \\ \forall x \in I \setminus \{x_0\}; g'(x) \neq 0; \end{array} \right.$

If:

$$\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = l$$

and:

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = l$$

So:

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = l$$